

◆ ALPHA TEAM 1.0 ◆
ENGINEERING ADMISSION
PREVIOUS YEAR QUESTIONS
PYQ BOOK



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Team Alpha

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2. MATRICES

MHT-CET 2021 PYQs

1. If $A(\alpha) = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$, then $[A^2(\alpha)]^{-1} =$

- (A) $A(\alpha)$
- (B) $A^2(\alpha)$
- (C) $A(-2\alpha)$
- (D) $A(2\alpha)$

2. If inverse of $\begin{bmatrix} 1 & 2 & x \\ 4 & -1 & 7 \\ 2 & 4 & -6 \end{bmatrix}$ does not exist, then $x =$

- (A) -3
- (B) 2
- (C) 3
- (D) 0

3. If $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 3 & 3 & -4 \end{bmatrix}$, $B = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$ and $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ such that $AX = B$, then the value of $x_1 + x_2 + x_3 =$

- (A) 4
- (B) 5
- (C) 6
- (D) 3

4. If $A^{-1} = \begin{bmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 5 & 5 \end{bmatrix}$, then $A =$

(A) $\begin{bmatrix} -5 & 20 & -2 \\ -1 & 3 & 0 \\ 3 & -11 & 1 \end{bmatrix}$

(B) $\begin{bmatrix} -5 & 20 & 2 \\ -1 & 3 & 0 \\ 3 & 11 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} -5 & 20 & 2 \\ 1 & 3 & 0 \\ 3 & 11 & -1 \end{bmatrix}$

(D) $\begin{bmatrix} -5 & 20 & -2 \\ 1 & 3 & 0 \\ 3 & 11 & 1 \end{bmatrix}$

5. If $A^{-1} = \begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$, then $(AB)^{-1} =$

(A) $\begin{bmatrix} 2 & 7 \\ 3 & -1 \end{bmatrix}$

(B) $\begin{bmatrix} 2 & -7 \\ -3 & 11 \end{bmatrix}$

(C) $\begin{bmatrix} 2 & -3 \\ -7 & 11 \end{bmatrix}$

(D) $\begin{bmatrix} 2 & 3 \\ 7 & -11 \end{bmatrix}$



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6. If $A = \begin{bmatrix} 3 & 2 & 4 \\ 1 & 2 & 1 \\ 3 & 2 & 6 \end{bmatrix}$ and A_{ij} are cofactors of the elements a_{ij} of A , then $a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$ is equal to

- (A) 8
- (B) 6
- (C) 4
- (D) 0

7. If $F(\alpha) = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$, where $\alpha \in \mathbb{R}$, then $[F(\alpha)]^{-1} =$

- (A) $F(-\alpha)$
- (B) $F(2\alpha)$
- (C) $F(\alpha)$
- (D) $F(3\alpha)$

8. If $A = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 1 & -2 \\ 0 & 2 & 1 \end{bmatrix}$, $\text{adj } A = \begin{bmatrix} 5 & x & -2 \\ 1 & 1 & 0 \\ -2 & -2 & y \end{bmatrix}$, then value of $x + y$ is

- (A) 6
- (B) 3
- (C) 4
- (D) -5

9. If $A^{-1} = \frac{1}{2} \begin{bmatrix} -1 & 1 & -4 \\ 2 & -1 & 2 \end{bmatrix}$, then $2A + I_2 =$, where I_2 is a unit matrix of order 2

(A) $\begin{bmatrix} 5 & 8 \\ 1 & 2 \end{bmatrix}$

(B) $\begin{bmatrix} 5 & 8 \\ 2 & 2 \end{bmatrix}$

(C) $\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$

(D) $\begin{bmatrix} 2 & 8 \\ 2 & 3 \end{bmatrix}$

10. If $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & a & 1 \end{bmatrix}$ and $A^{-1} = \frac{1}{2} \begin{bmatrix} 1 & -1 & 1 \\ -8 & 6 & 2c \\ 5 & -3 & 1 \end{bmatrix}$, then values of a and c are respectively

(A) $\frac{1}{2}, \frac{1}{2}$

(B) $-1, 1$

(C) $2, -\frac{1}{2}$

(D) $1, -1$

11. If $A = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then $\text{adj } A =$

(A) $\begin{bmatrix} -\cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(B) $\begin{bmatrix} \cos \theta & \sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(D) $\begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

12. If $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$, then $A^{-1} =$

(A) $\frac{1}{2} \begin{bmatrix} 0 & 1 & 2 \\ 3 & 2 & 1 \\ 4 & 2 & 3 \end{bmatrix}$

(B) $\frac{1}{2} \begin{bmatrix} -1 & 1 & -1 \\ -8 & 6 & -2 \\ 5 & -3 & 1 \end{bmatrix}$

(C) $\frac{1}{2} \begin{bmatrix} 5 & 2 & 3 \\ -1 & -6 & 2 \\ 1 & 2 & -1 \end{bmatrix}$

(D) $\frac{1}{2} \begin{bmatrix} -1 & -1 & -1 \\ -8 & 6 & -3 \\ 5 & 5 & 1 \end{bmatrix}$

13. If $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix}$, then the value of determinant of A^{-1} is

(A) -6

(B) $-\frac{1}{6}$

(C) $\frac{1}{36}$

(D) 36

14. If $A = \begin{bmatrix} 2 & k \\ -2 & -k \end{bmatrix}$, then A^{-1} does not exist if $k =$

(A) 3

(B) ± 2

(C) 0

(D) ± 1

15. If $A = \begin{bmatrix} 5a & -b \\ 3 & 2 \end{bmatrix}$ and $A \operatorname{adj} A = AA^T$, then $5a + b =$

(A) 13

(B) 4

(C) -1

(D) 5

16. For an invertible matrix A , if $A(\operatorname{adj} A) = \begin{bmatrix} 20 & 0 \\ 0 & 20 \end{bmatrix}$, then

$|A| =$

(A) -200

(B) 200

(C) -2

(D) 20

17. If $A = \begin{bmatrix} 1 & 2 & 1 \\ -1 & 1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ -3 & 1 \\ 0 & 2 \end{bmatrix}$, then $(AB)^{-1} =$

(A) $\begin{bmatrix} 5 & -6 \\ -4 & 5 \end{bmatrix}$

(B) $\begin{bmatrix} 5 & 6 \\ 4 & 5 \end{bmatrix}$

(C) $\begin{bmatrix} -5 & 6 \\ -4 & 5 \end{bmatrix}$

(D) $\begin{bmatrix} -5 & -6 \\ -4 & -5 \end{bmatrix}$

18. If $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$ and $A^{-1} = KA$, then K is

(A) 19

(B) $-\frac{1}{19}$

(C) -19

(D) $\frac{1}{19}$

9. If $A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 1 & 2 \\ 1 & 2 & 4 \end{bmatrix}$, then $A(\text{adj } A) =$

$$(A) \begin{bmatrix} -1/3 & 0 & 0 \\ 0 & -1/3 & 0 \\ 0 & 0 & -1/3 \end{bmatrix}$$

$$(B) \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$(C) \begin{bmatrix} 1 & 2 & 3 \\ -1 & 1 & 2 \\ 1 & 2 & 4 \end{bmatrix}$$

$$(D) \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & 2 \\ 3 & 2 & 4 \end{bmatrix}$$

20. Which of the following matrices are invertible?

$$A = \begin{bmatrix} 2 & 3 \\ 10 & 15 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 3 \\ 2 & 2 & 3 \end{bmatrix}, C = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \\ 4 & 6 & 8 \end{bmatrix}, D =$$

$$\begin{bmatrix} 2 & 4 & 2 \\ 1 & 1 & 0 \\ 1 & 4 & 5 \end{bmatrix}$$

(A) both A and B

(B) only C

(C) only A

(D) only D

21. If $A = \begin{bmatrix} 2 & -2 \\ -3 & -2 \end{bmatrix}$, $B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, then $(B^{-1}A^{-1})^{-1} =$

(A) $A = \begin{bmatrix} -2 & -2 \\ -3 & 2 \end{bmatrix}$

(B) $A = \begin{bmatrix} 2 & 2 \\ -2 & -3 \end{bmatrix}$

(C) $A = \begin{bmatrix} 3 & -2 \\ 2 & 2 \end{bmatrix}$

(D) $A = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$

22. If $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & a & 2 \\ 2 & 4 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} 13 & 2 & b \\ -3 & -1 & 2 \\ -2 & 0 & 1 \end{bmatrix}$ where matrix B is inverse of matrix A , then the value of a and b are

(A) $a = -5, b = 7$

(B) $a = 7, b = -5$

(C) $a = -7, b = 5$

(D) $a = 5, b = -7$

23. For a 3×3 matrix A , if $A(\text{adj } A) = \begin{bmatrix} -10 & 0 & 0 \\ 0 & -10 & 0 \\ 0 & 0 & -10 \end{bmatrix}$, then the value of determinant of A is

(A) 100

(B) -1000

(C) -10

(D) 20

24. If $A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$ and A^{-1} does not exist, then $\lambda =$
(where $i = \sqrt{-1}$)

(A) ± 2

(B) ± 1

(C) 0

(D) ± 3

25. If $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \\ 1 & 2 & 4 \end{bmatrix}$, and $A(\text{adj } A) = kI$, then the value of
 $(k + 1)^4$ is

(A) 256

(B) 81

(C) 16

(D) 625

26. If $AX = B$, where $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $B = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$,
then $2x + y - z =$

(A) 2

(B) 1

(C) 4

(D) -2

◆ MHT-CET 2022 PYQs

1. If $\begin{bmatrix} 1 & 1 & 1 \\ 1 & -2 & -2 \\ 1 & 3 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 4 \end{bmatrix}$, then $2x - y + z =$

- (A) 1
- (B) 2
- (C) 3
- (D) -3

2. If $\begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 2 & -3 \\ 5 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then $A =$

- (A) $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$
- (B) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$
- (C) $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$
- (D) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

3. If $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & -3 \\ -1 & 2 & 3 \end{bmatrix}$, then $A_{31} + A_{32} + A_{33} =$ where A_{ij} is cofactor of a_{ij} , where $A = [a_{ij}]_{3 \times 3}$

- (A) 1
- (B) 11
- (C) 10

(D) 0

4. If matrix $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$ is such that $AX = I$, where I is 2×2 unit matrix, then $X =$

(A) $\frac{1}{5} \begin{bmatrix} -3 & 2 \\ 4 & -1 \end{bmatrix}$

(B) $\frac{1}{5} \begin{bmatrix} 3 & 2 \\ 4 & 1 \end{bmatrix}$

(C) $\frac{1}{5} \begin{bmatrix} 3 & -2 \\ -4 & 1 \end{bmatrix}$

(D) $\frac{1}{5} \begin{bmatrix} 1 & -3 \\ -4 & -1 \end{bmatrix}$

5. If $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ such that $A^2 - 4A + 3I = 0$, where I is a unit matrix of order 2, then A^{-1} is

(A) $\frac{1}{3} \begin{bmatrix} -1 & 2 \\ 1 & -2 \end{bmatrix}$

(B) $\frac{1}{3} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

(C) $\frac{1}{3} \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$

(D) $\frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$



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6. If $A = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & -3 \\ 2 & 1 & 0 \end{bmatrix}$ and $B = \text{adj } A$, $C = 5A$, then $\frac{|\text{adj } B|}{|C|} =$

(A) -1

(B) 25

(C) 5

(D) 1

7. If $A = [a_{ij}]_{3 \times 3} = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 5 \\ 2 & 4 & 7 \end{bmatrix}$ and A_{ij} is a cofactor of a_{ij} , then $a_{11}A_{21} + a_{12}A_{22} + a_{13}A_{23}$ equal to

(A) -1

(B) 0

(C) 1

(D) 2

8. The element in the third row and second column of the inverse of the matrix $\begin{bmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{bmatrix}$ is

(A) -2

(B) 0

(C) 1

(D) 2

9. If $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 4 \\ 3 & 4 & 3 \end{bmatrix}$, then $A^{-1} =$

(A) $\frac{1}{4} \begin{bmatrix} -7 & 6 & -1 \\ 9 & -6 & -1 \\ -5 & 2 & 1 \end{bmatrix}$

(B) $\frac{1}{4} \begin{bmatrix} 7 & 6 & 1 \\ 9 & -1 & 1 \\ -5 & 2 & 1 \end{bmatrix}$

(C) $\frac{1}{4} \begin{bmatrix} -7 & 6 & -1 \\ 9 & -6 & 2 \\ -5 & 2 & 1 \end{bmatrix}$

(D) $\frac{1}{4} \begin{bmatrix} -7 & -6 & -1 \\ 9 & 6 & -1 \\ -5 & -2 & 1 \end{bmatrix}$

10. If $A = \begin{bmatrix} 1+2i & i-2 \\ 0 & K \\ -1-2i & 0 \\ 2-i & 7 \end{bmatrix}$ and A^{-1} does not exist, then

$K =$ (where $i = \sqrt{-1}$)

(A) 7

(B) -7

(C) $1 + 2i$

(D) $1 - 2i$

11. If $A = \begin{bmatrix} 1 & 3 & -2 \\ -3 & 0 & -5 \\ 2 & 5 & 0 \end{bmatrix}$ and $A(\text{adj } A) = kI$, then the value
(where I is unit matrix of order 3)

(A) -25

(B) 25

(C) 85

(D) -85

12. If $|A| = -3$ and $A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1/3 & 0 \\ 3 & 2/3 & -1 \end{bmatrix}$, then $(\text{adj } A)$ is

(A) $\begin{bmatrix} -1/3 & 0 & 0 \\ -1/3 & -1/9 & 2/9 \\ 1 & -1/9 & 9/3 \end{bmatrix}$

(B) $\begin{bmatrix} 1 & 0 & 0 \\ -1 & 1/9 & 2/9 \\ -1 & 2/9 & 9/3 \end{bmatrix}$

(C) $\begin{bmatrix} 3 & 0 & 0 \\ -3 & -1 & 0 \\ 9 & 2 & -3 \end{bmatrix}$

(D) $\begin{bmatrix} -3 & 0 & 0 \\ 3 & -1 & 0 \\ -9 & -2 & 3 \end{bmatrix}$

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13. If matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$ and $A^{-1} = \alpha I + \beta A$ where I is a unit matrix of order 2 and α, β are constants, then the value of $\alpha + \beta + \alpha\beta$ is

- (A) -11
- (B) 11
- (C) 7
- (D) -7

14. If $(BA)^{-1} = C$ where $B = \begin{bmatrix} 2 & 6 & 4 \\ 1 & 0 & 1 \\ -1 & 1 & -1 \end{bmatrix}$ and $A^{-1} = C$ then

(A) $\begin{bmatrix} -1 & 0 & 1 \\ 1 & 1 & 3 \\ 2 & 0 & 2 \end{bmatrix}$

(B) $\begin{bmatrix} -3 & -5 & -5 \\ -3 & 0 & 9 & 2 \\ 2 & 14 & 6 \end{bmatrix}$

(C) $\begin{bmatrix} -3 & 5 & -5 \\ -3 & 0 & 9 \\ 2 & 14 & 6 \end{bmatrix}$

(D) $\begin{bmatrix} -3 & 5 & 5 \\ -3 & 0 & 9 \\ 2 & 2 & 6 \end{bmatrix}$

15. If A is non-singular matrix of order 3 such that $(A - 2I)(A - 4I) = 0$, then $\frac{1}{6}A + \frac{4}{3}A^{-1}$ is

(where I is a unit matrix of order 3 and 0 is a null matrix of order 3)

- (A) 0
- (B) I
- (C) $2I$
- (D) $6I$

16. If $A = \begin{bmatrix} 2 & -3 \\ 4 & 1 \end{bmatrix}$, then $A + \text{adj } A$ is

- (A) $\begin{bmatrix} 1 & -3 \\ 4 & 2 \end{bmatrix}$
- (B) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- (C) $\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$
- (D) $\begin{bmatrix} 1 & 3 \\ -4 & 2 \end{bmatrix}$

17. If $A = [a_{ij}]_{3 \times 3} = \begin{bmatrix} 1 & 3 & 3 \\ -1 & 2 & 2 \\ 1 & 1 & 4 \end{bmatrix}$ and A_{ij} is a cofactor of a_{ij} , then the value of $a_{31}A_{31} + a_{32}A_{32} + a_{33}A_{33}$ is equal to

- (A) 0
- (B) 20
- (C) 5
- (D) 15

18. If $A = \begin{bmatrix} 2 & -3 \\ -4 & 1 \end{bmatrix}$, then $\text{adj}(3A^2 + 12A)$ is equal to

(A) $\begin{bmatrix} 72 & -84 \\ -63 & 51 \end{bmatrix}$

(B) $\begin{bmatrix} 51 & 63 \\ 84 & 72 \end{bmatrix}$

(C) $\begin{bmatrix} 51 & 84 \\ 63 & 72 \end{bmatrix}$

(D) $\begin{bmatrix} 72 & -63 \\ -84 & 51 \end{bmatrix}$

19. If $A = \begin{bmatrix} 1 & a & 3 \\ 1 & 1 & 5 \\ 2 & 4 & 7 \end{bmatrix}$ and $A^{-1} = \begin{bmatrix} 13 & 2 & -7 \\ -3 & b & 2 \\ -2 & 0 & 1 \end{bmatrix}$, then the values of a and b respectively

(A) 2, 1

(B) 2, -1

(C) -1, 2

(D) 1, 2

20. If $A = [a_{ij}]_{3 \times 3} = \begin{bmatrix} 3 & 2 & 4 \\ 1 & 4 & 1 \\ 2 & 6 & 3 \end{bmatrix}$ and A_{ij} is a cofactor of a_{ij} , then the value of $a_{21}A_{21} + a_{22}A_{22} + a_{23}A_{23}$ is equal to

(A) 18

(B) -8

(C) 8

(D) 0

21. If $A = \begin{bmatrix} 2 & 3 \\ -4 & 1 \end{bmatrix}$, then $\text{adj}(3A^2 + 12A)$ is equal to

(A) $\begin{bmatrix} -21 & 63 \\ 84 & 0 \end{bmatrix}$

(B) $\begin{bmatrix} -21 & -63 \\ 84 & 0 \end{bmatrix}$

(C) $\begin{bmatrix} 21 & -63 \\ 84 & 0 \end{bmatrix}$

(D) $\begin{bmatrix} 21 & 63 \\ 84 & 0 \end{bmatrix}$

◆ MHT-CET 2023 PYQs

1. If $A = \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix}$, then the inverse of $(2A^2 + 5A)$ is

(A) $\frac{1}{95} \begin{bmatrix} 7 & 3 \\ 3 & 4 \end{bmatrix}$

(B) $\frac{1}{95} \begin{bmatrix} -7 & 3 \\ 3 & -4 \end{bmatrix}$

(C) $\frac{1}{95} \begin{bmatrix} -7 & -3 \\ 3 & 4 \end{bmatrix}$

(D) $\frac{1}{95} \begin{bmatrix} 4 & 3 \\ 3 & 7 \end{bmatrix}$



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2. If $\begin{bmatrix} \cos(A+B) & -\sin(A+B) \\ \sin A & \cos A \\ \sin A & \cos A \end{bmatrix} \begin{bmatrix} \cos B & \sin B \\ \sin B & \cos B \end{bmatrix} = 0$, then the value of B

(A) $n\pi, n \in \mathbb{Z}$

(B) $(2n + 1)\frac{\pi}{2}, n \in \mathbb{Z}$

(C) $(2n + 1)\frac{\pi}{4}, n \in \mathbb{Z}$

(D) $2n\frac{\pi}{3}, n \in \mathbb{Z}$

3. If $B = \begin{bmatrix} 1 & \alpha & 2 \\ 1 & 2 & 2 \\ 2 & 3 & 3 \end{bmatrix}$ is the adjoint of a 3×3 matrix A and $|A| = 5$, then α is equal to

(A) 25

(B) 27

(C) $3\sqrt{3}$

(D) 5

4. If $B = \begin{bmatrix} 3 & \alpha & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$ is the adjoint of a 3×3 matrix A and $|A| = 4$, then α is equal to

(A) 1

(B) 0

(C) -1

(D) -2

5. Let $\omega \neq 1$ be a cube root of unity and S be the set all non-singular matrices of the form $\begin{bmatrix} 1 & a & b \\ \omega & 1 & c \\ \omega^2 & \omega & 1 \end{bmatrix}$, where each of a, b and c is either ω or ω^2 , then the number of distinct matrices in the set S is

(A) 2

(B) 6

(C) 4

(D) 8

6. If $P = \begin{bmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{bmatrix}$ is the adjoint of a 3×3 matrix A and $|A| = 4$, then value of α is

(A) 4

(B) 11

(C) 5

(D) 0

7. If $A = \begin{bmatrix} i & 1 \\ 1 & 0 \end{bmatrix}$ where $i = \sqrt{-1}$ and $B = A^{2029}$, then $B^{-1} =$

(A) $-A$

(B) $\text{adj } A$

(C) $-I$

(D) $-\text{adj } A$

8. If the matrix $A = \begin{bmatrix} 1 & 2 \\ -5 & 1 \end{bmatrix}$ and $A^{-1} = xA + yI$, where I is a unit matrix of order 2, then the value of $2x + 3y$ is

(A) $\frac{8}{11}$

(B) $\frac{4}{11}$

(C) $-\frac{8}{11}$

(D) $-\frac{4}{11}$

9. If $A = \begin{bmatrix} 2a & -3b \\ 3 & 2 \end{bmatrix}$ and $A \text{ adj } A = AA^T$, then $2a + 3b$ is

(A) -1

(B) 1

(C) 5

(D) -5

10. If $A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 1 & 2 \\ 1 & 2 & 4 \end{bmatrix}$ and A_{ij} is a cofactor of a_{ij} , then the value of $a_{21}A_{21} + a_{22}A_{22} + a_{23}A_{23}$ is

(A) 0

(B) -2

(C) 4

22. If $A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ 1 & 2 \\ 1 & 2 \end{bmatrix}$, then $(AB)^{-1} =$

(A) $\begin{bmatrix} -17/5 & 9/5 \\ 2 & -1 \end{bmatrix}$

(B) $\begin{bmatrix} 17/9 & 2 \\ 5/5 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} -17/5 & 2 \\ -9/5 & 5 \end{bmatrix}$

(D) $\begin{bmatrix} 17/5 & 9/5 \\ 2 & 1 \end{bmatrix}$

23. Given $A = \begin{bmatrix} x & 3 & 2 \\ 1 & y & 4 \\ 2 & 2 & z \end{bmatrix}$, if $xyz = 60$ and $8x + 4y + 3z = 20$, then $A(\text{adj } A)$ is equal to

(A) $\begin{bmatrix} 108 & 0 & 0 \\ 0 & 108 & 0 \\ 0 & 0 & 108 \end{bmatrix}$

(B) $\begin{bmatrix} 60 & 0 & 0 \\ 0 & 60 & 0 \\ 0 & 0 & 60 \end{bmatrix}$

(C) $\begin{bmatrix} 68 & 0 & 0 \\ 0 & 68 & 0 \\ 0 & 0 & 68 \end{bmatrix}$

$$(D) \begin{bmatrix} 20 & 0 & 0 \\ 0 & 20 & 0 \\ 0 & 0 & 20 \end{bmatrix}$$

◆ MHT-CET 2024 PYQs

01. If $A = \begin{bmatrix} 3 & -1 \\ -4 & 2 \end{bmatrix}$, then A^{-1} is

(A) $\frac{1}{2} \begin{bmatrix} 2 & 3 \\ 3/2 & 2 \end{bmatrix}$

(B) $\frac{1}{2} \begin{bmatrix} 1 & 1/2 \\ 3/2 & 2 \end{bmatrix}$

(C) $\frac{1}{2} \begin{bmatrix} 1 & -1/2 \\ 3/2 & -2 \end{bmatrix}$

(D) $\frac{1}{2} \begin{bmatrix} 1 & 1/2 \\ 3/2 & 2 \end{bmatrix}$

02. Let $X = b$, $A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 0 & 1 \\ 3 & 2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}$

If $AX = B$, then the value of $2a - 3b + 4c$ will be

(A) 0

(B) -4

(C) 6

(D) 4

03. If $A = \begin{bmatrix} 2 & -2 \\ 4 & 3 \end{bmatrix}$, then $A^{-1} =$

(A) $\frac{1}{14} \begin{bmatrix} 3 & 2 \\ 2 & -4 \end{bmatrix}$

(B) $\frac{1}{14} \begin{bmatrix} 3 & 2 \\ -4 & 2 \end{bmatrix}$

(C) $\frac{1}{14} \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix}$

(D) $\frac{1}{14} \frac{1}{4} \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix}$

04. For the system $x - y + z = 4, 2x + y - 3z = 0, x + y + z = 2$, the values of x, y, z respectively are given by

(A) 2, 1, 1

(B) 2, 1, -1

(C) 2, -1, 1

(D) -2, 1, 1

05. Suppose A is any 3×3 non-singular matrix and $(A - 3I)(A - 5I) = 0$ where $I = I_3$ and $O = O_3$. Here O_3 represent zero matrix of order 3 and I_3 is an identity matrix order 3. If $\alpha A + \beta A^{-1} = 4I$, then $\alpha + \beta$ is equal to

(A) 13

(B) 7

(C) 12

(D) 8

06. Let $A = \begin{bmatrix} 1 & 2 \\ -5 & 1 \end{bmatrix}$ and $A^{-1} = xA + yI_2$ (where I_2 is unit matrix of order 2), then

(A) $x = -\frac{1}{11}, y = \frac{2}{11}$

(B) $x = \frac{1}{11}, y = \frac{2}{11}$

(C) $x = \frac{1}{11}, y = -\frac{2}{11}$

(D) $x = \frac{1}{11}, y = -\frac{2}{11}$

07. If $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & 3 \\ 3 & 2 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 0 & b \\ 7 & -1 & -2 \\ c & 1 & 1 \end{bmatrix}$ and if matrix B is the inverse of matrix A , then the value of $4a + 2b - c$ is

(A) 6

(B) -14

(C) 14

(D) -6

08. For the matrix $A = \begin{bmatrix} 2 & 0 & -1 \\ 3 & 1 & 2 \\ -1 & 1 & 2 \end{bmatrix}$, then the matrix of cofactors is

(A) $\begin{bmatrix} 0 & 8 & -4 \\ -1 & 3 & 2 \\ 1 & -7 & 2 \end{bmatrix}$

(B) $\begin{bmatrix} 0 & -8 & 4 \\ -1 & -3 & 2 \\ 1 & -7 & 2 \end{bmatrix}$

$$(C) \begin{bmatrix} 0 & 8 & -4 \\ 1 & -3 & 2 \\ -1 & 7 & -2 \end{bmatrix}$$

$$(D) \begin{bmatrix} 0 & -8 & 4 \\ 0 & 1 & 3 & 2 \\ -1 & -7 & 2 \end{bmatrix}$$

09. If $A + B = \begin{bmatrix} 1 & \tan \frac{\theta}{2} \\ -\tan \frac{\theta}{2} & 1 \end{bmatrix}$ where A is symmetric and B is skew-symmetric matrix, then the matrix $(A^{-1}B + AB^{-1})$ at $\theta = \frac{\pi}{6}$ is given by

$$(A) \begin{bmatrix} 1 & 2\sqrt{3} \\ 2\sqrt{3} & 1 \end{bmatrix}$$

$$(B) \begin{bmatrix} -1 & -2\sqrt{3} \\ 2\sqrt{3} & 1 \end{bmatrix}$$

$$(C) \begin{bmatrix} 0 & 2\sqrt{3} \\ 2\sqrt{3} & 0 \end{bmatrix}$$

$$(D) \begin{bmatrix} 0 & -2\sqrt{3} \\ 2\sqrt{3} & 0 \end{bmatrix}$$

10. Inverse of the matrix $\begin{bmatrix} 0.8 & -0.6 \\ 0.6 & 0.8 \end{bmatrix}$ is

$$(A) \begin{bmatrix} 0.8 & 0.6 \\ -0.6 & 0.8 \end{bmatrix}$$

$$(B) \begin{bmatrix} -0.8 & 0.6 \\ -0.6 & 0.8 \end{bmatrix}$$

$$(C) \begin{bmatrix} -0.8 & -0.6 \\ 0.6 & 0.8 \end{bmatrix}$$

(D) $\begin{bmatrix} 8 & -6 \\ 6 & 8 \end{bmatrix}$

11. Let $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$ such that $AX = B$, then

$X =$

(A) $\begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$

(B) $\begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$

(C) $\begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$

(D) $\begin{bmatrix} -2 \\ 1 \\ -1 \end{bmatrix}$

12. If $A = \begin{bmatrix} 2 & 1 \\ 7 & 4 \end{bmatrix}$, then $(A^2 - 5A)^{-1}$ is

(A) $\frac{1}{4} \begin{bmatrix} -3 & 7 \\ 7 & -1 \end{bmatrix}$

(B) $\frac{1}{4} \begin{bmatrix} -3 & -7 \\ 7 & -1 \end{bmatrix}$

(C) $\frac{1}{4} \begin{bmatrix} 3 & 7 \\ 7 & 1 \end{bmatrix}$

(D) $\frac{1}{4} \begin{bmatrix} 3 & -7 \\ 7 & -1 \end{bmatrix}$

13. Let A and B be 3×3 real matrices such that A is symmetric matrix and B is skew-symmetric matrix. Then the system of

linear equations $(A^2B^2 - B^2A^2)X = 0$ where X is 3×1 column matrix of unknown variables and 0 is a 3×1 null matrix, has

- (A) a unique solution
- (B) exactly two solutions
- (C) no solution
- (D) infinitely many solutions

14. If $A = \begin{bmatrix} 5a & -b \\ 3 & 2 \end{bmatrix}$ and $A \cdot \text{adj } A = AA^T$, then $5a + b$ is equal to

- (A) -1
- (B) 5
- (C) 3
- (D) 13

15. Let $A = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}$, $x \in \mathbb{R}^+$ and $A^4 = [a_{ij}]$, If $a_{11} = 109$, then $(A^4)^{-1} =$

- (A) $\begin{bmatrix} 109 & 33 \\ 33 & 10 \end{bmatrix}$
- (B) $\begin{bmatrix} 10 & 33 \\ 33 & 10 \end{bmatrix}$
- (C) $\begin{bmatrix} 10 & 33 \\ 33 & 109 \end{bmatrix}$

(D) $\frac{1}{10} \begin{bmatrix} 10 & -33 \\ -33 & 109 \end{bmatrix}$



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16. Let $A = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix}$ and $A^{-1} = \alpha I + \beta A, \alpha, \beta \in \mathbb{R}, I$ is the identity matrix of order 2, then $4(\alpha - \beta)$ is

(A) $\frac{8}{3}$

(B) 4

(C) 2

(D) 5

 **Answer Keys (उत्तरतालिका)**

 **MHT-CET 2021 Answer Key**

Q.N o.	An s	Q.N o.	An s	Q.N o.	An s	Q.N o.	An s	Q.N o.	An s
1	C	2	A	3	D	4	A	5	C
6	A	7	A	8	D	9	D	10	D
11	C	12	B	13	B	14	B	15	D
16	D	17	C	18	D	19	B	20	D
21	A	22	D	23	C	24	B	25	A
26	A								

 **MHT-CET 2022 Answer Key**

Q.N o.	An s	Q.N o.	An s	Q.N o.	An s	Q.N o.	An s	Q.N o.	An s
1	D	2	A	3	D	4	A	5	C
6	D	7	B	8	A	9	C	10	B
11	B	12	D	13	A	14	B	15	B
16	C	17	D	18	B	19	B	20	C
21	B	22	A	23	C				

🔴 MHT-CET 2023 Answer Key

Q.N o.	An s	Q.N o.	An s	Q.N o.	An s	Q.N o.	An s	Q.N o.	An s
1	A	2	B	3	B	4	A	5	A
6	B	7	D	8	B	9	C	10	D
11	B	12	D	13	B				

MHT-CET 2024 Answer Key

Q.N o.	An s	Q.N o.	An s	Q.N o.	An s	Q.N o.	An s	Q.N o.	An s
1	D	2	D	3	B	4	C	5	D
6	A	7	C	8	B	9	D	10	A
11	B	12	C	13	D	14	B	15	D
16	B								

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